

SMART INDIA HACKATHON 2026



VERCEL

- Problem Statement ID – 26179
- Problem Statement Title – AI Powered Retail Intelligence Platform
- Theme – Miscellaneous
- PS Category – Hardware
- Team ID –
- Team Name (Registered on portal) –



IDEA TITLE

Stores currently **lack** the ability to track empty shelves, customer movement, and long queues in real time.



- **Poor staffing decisions** increase operating costs.
- Stores have **limited visibility** into shopper behavior.
- Manual monitoring **wastes staff time**.
- **Weak connectivity** makes cloud systems unreliable.
- **Privacy concerns** limit customer tracking.

*Prototype

Detailed Report*

[Video Website](#)

[Report GitHub](#)

PROPOSED SOLUTION

- Uses **existing store** cameras and **processes data locally**.
- Detects **empty shelves** and long queues in **real time**.
- Gives **quick, privacy-safe** alerts to staff.
- Helps **improve daily store operations**.
- Works with **existing** POS/ERP systems.



Edge AI



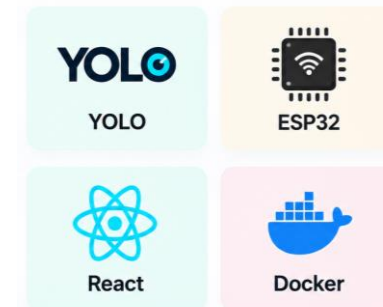
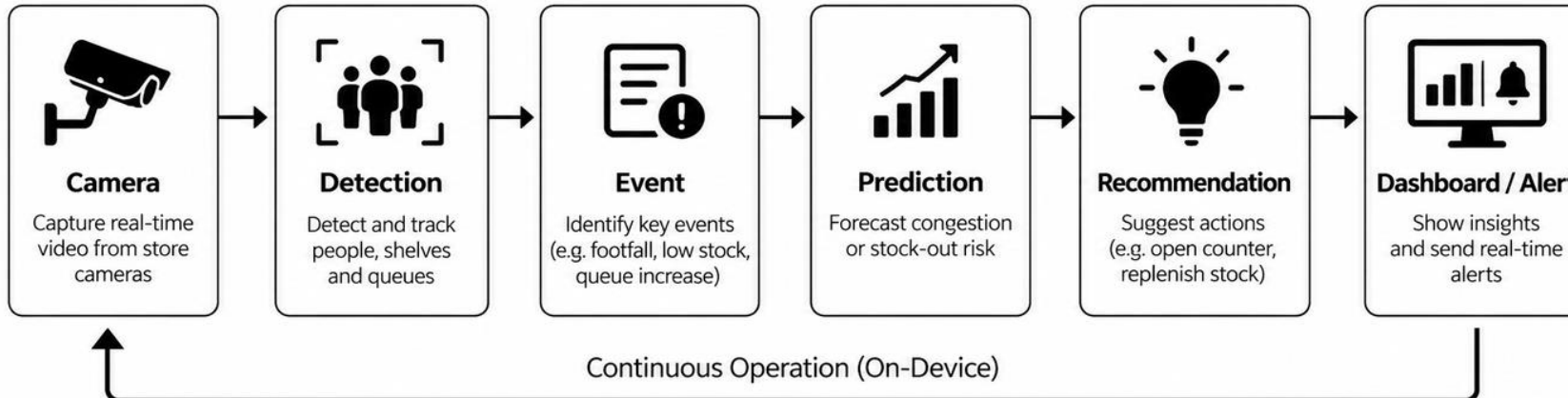
Real-Time Insights



Privacy-First

Vercel

TECHNICAL APPROACH



1. Camera Inputs

Multiple views, multiple insights



2. Stream Manager

- ✓ Ingest & manage multiple video streams (RTSP / IP)
- ✓ Handles buffering, synchronization
- ✓ Fault tolerance

3. Inference Scheduler

- ✓ Prioritize streams and allocate compute resources
- ✓ Adaptive scheduling based on load and camera importance
- ✓ Manages FPS per stream

4. Computer Vision

- ✓ Detection, Tracking & Analysis (YOLO + OpenCV)
- ✓ People, objects, queues, shelves and movement understanding
- ✓ ROI-based processing

5. Anonymous Tracking

- ✓ Track people & objects without storing identity
- ✓ Maintain privacy with on-device processing and ID anonymization
- ✓ Trajectory, dwell time and zone analytics

6. Event Engine

- ✓ Generate events (Footfall, Queue, Low Stock, Dwell Time, etc.)
- ✓ Send to analytics, recommendation and local storage
- ✓ Real-time event processing

7. Prediction

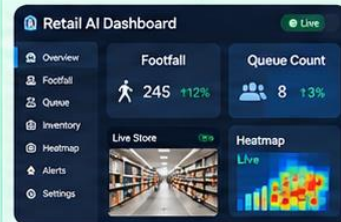
- ✓ Forecast congestion, demand and stock-out risk (ML Models)
- ✓ Trend analysis & insights

8. Recommendation

- ✓ Suggest actions (Open counter, Replenish stock, etc.)
- ✓ Optimize staff allocation

9. Dashboard + Alerts

- ✓ Real-time insights (SMS / Dashboard)
- ✓ Live metrics, heatmaps, alerts and reports.



Local Event Store

- ✓ Store events locally (SQLite / PostgreSQL)
- ✓ Works offline (on-device storage)

Internet Available

Optional Cloud Sync

- ✓ Aggregated data & analytics (reports, long-term insights)
- ✓ Model updates (optional)

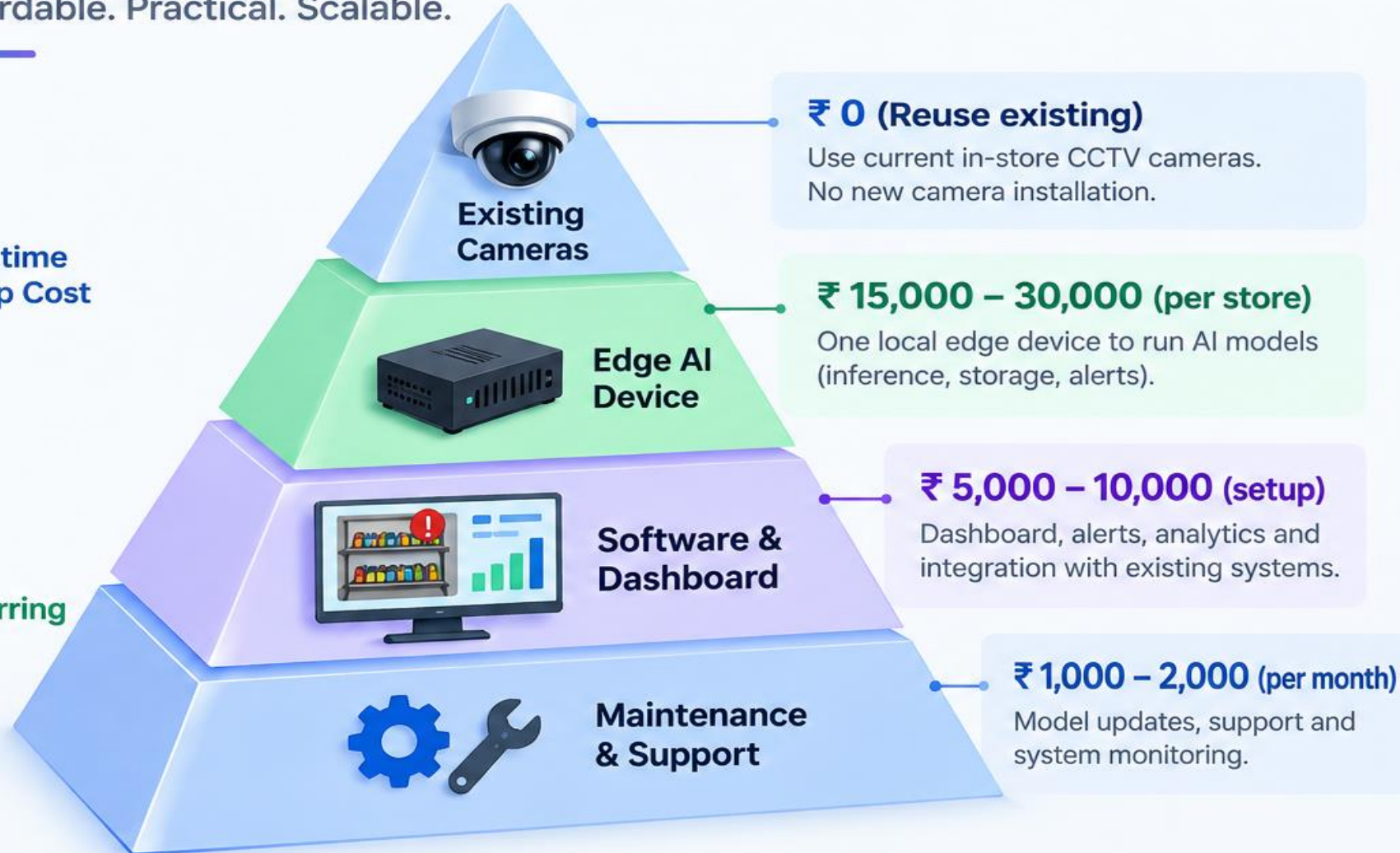
Continuous Improvement

Insights used to optimize scheduling, models and operations

Affordable. Practical. Scalable.

One-time
Setup Cost

Low
Recurring
Cost



Total Cost
(per store)

~₹ 25K – 40K
(one-time setup)

~₹ 1K – 2K
(monthly)



Low cost, high impact
using existing infrastructure

Reactive Operations

Manual monitoring, delayed response, operational inefficiencies.



Stock-out on shelf

Long queues at checkout

Manual monitoring by staff



Predictive Operations

Real-time insights, proactive actions, smoother operations.



Shelf availability monitoring

Real-time dashboard & alerts

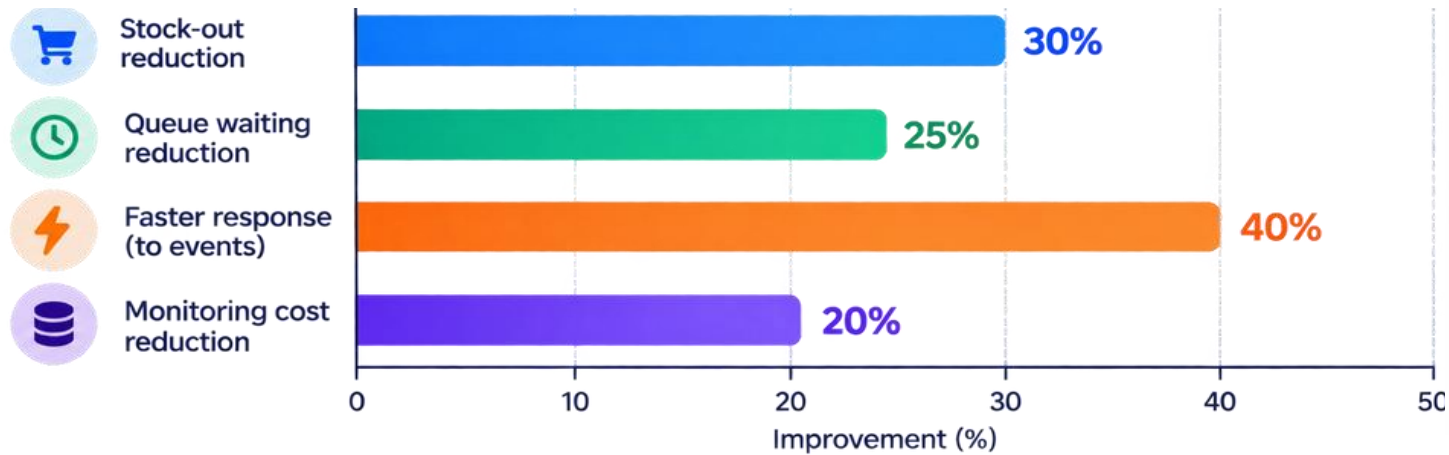
Optimized staff allocation

Shelves OK 98%

Queue Count 2

Footfall (Live)

No Critical Alerts



RESEARCH AND REFERENCES

Vercel

Computer Vision



YOLO-based people, queue and shelf detection.

Detects customers, queues, products and abnormal activity.

[Docs Link](#)

Retail Analytics



Footfall, dwell time and heatmaps.

Analyzes customer movement, popular zones and store engagement.

[Inspiration](#)

Queue Monitoring



Queue length, waiting time and service activity.

Monitors real-time queues and estimates waiting time.

[Docs Link](#)

Shelf Monitoring



Shelf availability and low-stock detection.

Detects empty or low-stock shelves using vision AI.

[Dataset Link](#)

Edge AI



On-device inference and privacy-preserving analytics.

Runs AI models at the edge with low latency and no video storage.

[Qualcomm](#)

Predictive Operations



Demand, congestion and stock-out risk.

Forecasts demand, predicts congestion and prevents stock-outs using ML.

[Paper Link](#)

Research Gap & Opportunity



Multimodal fusion, adaptive inference, low-cost deployment

Combine vision, sensor and transactional data for smarter, scalable retail intelligence.

[Paper Link](#)

UI/UX

Real-Time Insights
for Smarter Stores

Monitor.
Analyze.
Optimize.

